

Cross-Sector Extraction after the Standalone Standard Model Derivation

A Ten-Paper Research-Arc Synthesis from Charged-Lepton Mass Ratios to the Standard Model Charge-Lattice Ratio Class

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Abstract

This paper synthesizes a ten-paper cross-sector extraction research arc developed downstream of the standalone Admissibility And Standing Constraint (AASC) derivation of the Standard Model structural tuple. The earlier structural spine fixes the admissible Standard Model envelope: gauge-product structure, representation-response roles, Higgs/bridge fixation, Yukawa-realizability status, charge/anomaly/vacuum compatibility, mass-response slots, and the distinction between structural closure and full numerical parameter-table closure. The present synthesis asks what becomes extractable after that envelope is fixed under AASC.

The ARC yields two central quotient-level successes. First, the charged-lepton sector supplies a relation-level mass-shape extraction, producing the charged-lepton ratio pattern

$$[1 : 206.7703159727 : 3477.4728371047]$$

prior to one-anchor MeV display. Second, the later cross-sector modules extract the fixed-scope Standard Model charge-lattice / hypercharge-ratio class

$$[Y_Q : Y_u : Y_d : Y_L : Y_e : Y_H]_{\text{norm,sign}} = [1 : 4 : -2 : -3 : -6 : 3]_{\text{norm,sign}},$$

with conventional charge presentation $Q(u) = 2/3$, $Q(d) = -1/3$, $Q(\nu) = 0$, and $Q(e) = -1$. These are reported as quotient-level extractions, not as fitted numerical coincidences or as imports from the empirical parameter table.

The synthesis also records the supporting structural layer: charge-witness rigidity, Abelian charge normalization, anomaly-standing obstruction, anomaly-ledger compression, anomaly–Yukawa equivalence, Yukawa topology readiness, and Higgs-like bridge carrier rigidity. It records the negative and deferred frontier: zero-point vacuum-energy quotient-descent obstruction, neutrino boundary-kernel / invariant-witness reconstruction without PMNS numerical closure, full-network nondefinition as a blocked master-route result, and explicit deferral of gauge-coupling values, the fine-structure constant, Higgs/electroweak scale, numerical Yukawa eigenvalues, CKM/PMNS numerical values, exact charged-lepton pole masses, zero-anchor mass scale, cosmological constant value, and unrestricted larger-domain BSM exclusion.

The result is not a second derivation of the entire Standard Model and not a claim of complete empirical parameter-table closure. It is a synthesis of a ten-paper extraction program showing that, once the standalone Standard Model structure is fixed, AASC can isolate nontrivial quotient-level numerical and structural ratio objects while blocking unlicensed promotion of parameters, calibrations, and unresolved scale choices.

Keywords: AASC; reportability; charged leptons; charge lattice; hypercharge; zero-point energy; neutrino oscillation; quotient descent; Standard Model; numerical readiness.

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1 Introduction: why a synthesis paper is needed

1.1 The cross-sector target

A constraint formalism gains evidential force when it isolates more than one independent quotient target without target leakage. The Cross-Sector Singleton ARC tests whether the same admissibility discipline can extract, certify, obstruct, or defer targets across distinct physical sectors. The target was not simply

charged leptons only,

but cross-sector reportability:

mass-shape + charge-lattice + vacuum-offset obstruction + neutrino invariant reconstruction.

This synthesis is necessary because the ARC's outputs are distributed across admission papers, screening papers, extraction papers, obstruction papers, and final registries. Read separately, the papers preserve audit rigor. Read together, they form a proof spine for what AASC currently licenses and what it does not yet license.

1.2 Relation to the standalone Standard Model structural derivation

This synthesis begins after the standalone Standard Model structural tuple has been fixed by the prior AASC derivation [1]. That prior result supplies the admissible structural envelope: gauge-product structure, representation-response roles, Higgs/bridge fixation, Yukawa-realizability status, charge/anomaly/vacuum compatibility, mass-response slots, and the distinction between structural closure and full numerical parameter-table closure.

The present ARC does not rederive the entire Standard Model. It asks which targets become extractable, obstructed, structurally certified, blocked, or deferred once that envelope is fixed. In that downstream setting, the ARC reports two principal quotient-level extractions: the charged-lepton relation-level mass-shape ratio and the fixed-scope Standard Model charge-lattice / hypercharge-ratio class.

1.3 The problem of status laundering

The main failure mode in any cross-sector numerical program is status laundering: a local alignment, a normalized display, a structural carrier, or an empirical calibration is promoted into a stronger numerical claim than its certificates permit. The ARC therefore uses the following rule as a standing constraint:

structural compatibility $\nRightarrow$ numerical extraction.

Examples of forbidden promotions include

charge lattice $\nRightarrow \alpha$,

bridge carrier $\nRightarrow v$,

one-anchor charged-lepton display $\nRightarrow$ zero-anchor mass derivation,

and

PMNS invariant witness $\nRightarrow$ PMNS numerical value.

The value of the ARC is not only that it reports positive structures. It also reports the certified boundaries beyond which the present corpus does not license stronger claims.

1.4 What this synthesis contributes

CS7 provides the final registry. This synthesis paper converts that registry into a readable proof spine. It explains how the ARC moves from admissibility and standing discipline to two quotient-level extractions, several certified structural objects, two dedicated obstruction certificates, and a deferred numerical frontier.

This manuscript is a synthesis and reportability spine for the Cross-Sector ARC. It does not replace the primary derivations in CS1–CS7 and CS6A–CS6C. Its role is to consolidate their certified outputs, preserve scope boundaries, and state the dependency structure by which extracted, structural, obstruction, blocked, presentation-only, and deferred targets are separated.

1.5 Main conclusion

The Cross-Sector ARC establishes the following portfolio:

1. a charged-lepton relation-level mass-shape extraction;
2. a fixed-scope Standard-Model charge-lattice singleton extraction;
3. a zero-point vacuum-energy quotient-descent obstruction;
4. a neutrino boundary-kernel reconstruction and invariant witness certification;
5. a structural support layer involving charge witnesses, Abelian normalization, anomaly standing, anomaly/Yukawa compatibility, bridge carrier rigidity, and Yukawa topology readiness;
6. a deferred numerical frontier for coupling constants, Higgs/electroweak values, numerical Yukawa eigenvalues, exact pole masses, CKM/PMNS values, and cosmological values.

The ARC therefore does not claim full numerical completion of the Standard Model. It claims cross-sector quotient structure with explicit reportability boundaries.

2 Method: reportability under AASC

2.1 AASC

AASC denotes *Admissibility And Standing Constraint*. In the Cross-Sector ARC, AASC is not a new particle, force, or empirical model. It is the discipline used to decide whether a candidate quantity has standing as a target, a witness, a relation, a presentation, an obstruction, or a deferred object. A candidate can be written down before it has standing; the ARC is concerned with the additional conditions under which it becomes reportable.

2.2 Reportability classes

The synthesis uses the CS7 status taxonomy. These classes are mutually controlling: a target receives the strongest status actually certified for its declared scope, and no status is upgraded by proximity to stronger-looking neighbors.

Extracted. A quotient-level, relation-level, or value-class target is uniquely determined inside its declared scope. In this ARC the two principal extracted targets are

`ChargedLeptonRelationLevelAlgebraicExtracted`

and

`FixedScopeChargeLatticeSingletonExtracted.`

Certified structural. A non-numerical standing object, carrier, topology, invariant class, ledger, or reconstruction is certified. **Certified structural** is non-numerical and non-extractive. It is not a soft form of **Extracted**. It reports that a structure has standing, not that a numerical value has been derived.

Certified obstruction. A prediction-forming or identity-forming route is formally blocked. Examples are

`ZPEQuotientObstructionCertified`

and

`InteriorFlavorTotalizationObstructed.`

An obstruction certificate blocks a route. It does not automatically explain a target value.

Deferred. The target remains open because a necessary certificate is absent. Deferred targets are not failures of the whole framework; they are unresolved targets that cannot yet be reported as extracted.

Blocked. The route is not defined or cannot be evaluated under the current scope. The full cross-sector aggregate, for example, remains

`GlobalNetworkNotDefined.`

Presentation only. A displayed number, table, formula, normalized representative, or one-anchor display presents an extracted or certified object. It is not itself a primitive extraction unless the quotient-level target has been certified.

2.3 No-amplification rule

The no-amplification rule is the central reportability theorem of the ARC:

$Status(x) \leq$ the status permitted by the weakest unresolved blocker for x .

In practice:

`OneAnchorDisplay` $\not\Rightarrow$ `ZeroAnchorScale`,
`BridgeCarrierRigidityCertified` $\not\Rightarrow$ Higgs scale,
`ChargeLatticeExtraction` $\not\Rightarrow \alpha, e, g_i$.

No amount of compatible structure turns a deferred scale, a missing endpoint certificate, or an empirical anchor into a zero-anchor derivation.

2.4 L-metric and quotient rule

Metric values, scale values, pole values, coupling magnitudes, and empirical constants are licensed interior descriptions only when quotient-stable witness extraction is supplied. Otherwise they remain calibration, presentation, or deferred values. The rule is especially important for the ZPE and scale sectors: an apparently informative metric display can fail reportability if it does not descend through the relevant quotient.

3 ARC proof-spine overview

3.1 CS1–CS5: discipline before extraction

The early papers install the discipline that prevents later overclaiming. Table 1 summarizes the logic.

Table 1: Early Cross-Sector ARC function.

Stage	Function	Output
CS1	Candidate admission and claim-standing discipline	No target enters without a status ledger. Calibration and presentation cannot be mistaken for prediction.
CS2	Same-target spectral, residue, and mixing audit	Charged-lepton relation target aligned; cross-sector target drift blocked.
CS3	Charge, anomaly, and residual typing	Structural charge/anomaly envelope aligned; normalization and finite-audit routes controlled.
CS4	Hierarchy, bridge, and scale screening	No current absolute-scale closure; one-anchor and bridge-conditioned scale states remain bounded.
CS5	Global compatibility and no-amplification	Only the charged-lepton relation network entered singleton testing; the full network remained undefined.

3.2 CS5.5: non-CL target search

CS5.5 is the pivot away from a charged-lepton-only arc. It asks which non-CL candidates are extraction-ready, obstruction-ready, structural-only, or deferred under L-metric ratio-candidacy. It selected the fixed-scope Standard-Model charge-lattice target

$$Q_{SM}^{lat}$$

as the strongest non-CL extraction route. It also retained ZPE obstruction and neutrino invariant reconstruction as secondary non-CL branches.

3.3 CS6A–CS6C: extraction, obstruction, reconstruction

The CS6 branch produced three distinct result types:

- CS6A: positive extraction of the fixed-scope Standard-Model charge-lattice / hypercharge-ratio singleton;
- CS6B: obstruction certification for the absolute homogeneous zero-point vacuum-energy offset;
- CS6C: boundary-kernel reconstruction and invariant-witness certification in the declared finite PMNS scope.

These branches give the ARC a portfolio rather than a single isolated success.

3.4 CS7: final registry

CS7 assigns every ARC target a final reportability status. This synthesis retains that registry but frames it as a proof spine: the ARC extracted, certified, obstructed, deferred, and blocked exactly what its certificates permit.

4 Result I: charged-lepton relation-level mass-shape extraction

4.1 Target

The charged-lepton extraction target is the relation-level mass-shape class

$$[q_e^2 : q_\mu^2 : q_\tau^2].$$

It is not an absolute mass scale and not an exact pole-mass target.

4.2 Extracted ratio

The ARC reports the charged-lepton mass-shape ratio

$$[1 : 206.7703159727 : 3477.4728371047].$$

This is the relation-level object. The electron is used as the baseline coordinate in this presentation.

4.3 One-anchor display

When the empirical electron mass is declared as a scale anchor, the relation-level result can be displayed as

$$m_i^{(e)} = m_e^{obs} \frac{q_i^2}{q_e^2}.$$

This gives the one-anchor metric display

$$m_\mu \approx 105.6594 \text{ MeV},$$

$$m_\tau \approx 1776.9850 \text{ MeV}.$$

The metric display is legitimate only with the declared electron anchor.

4.4 Status

The final CS7 status is

`ChargedLeptonRelationLevelAlgebraicExtracted.`

The optional MeV display has status

`OneAnchorMetricDisplay.`

4.5 Boundary

The charged-lepton result does not derive exact pole masses, zero-anchor scale, the electron mass, the protected trace remainder, or a complete lepton-sector mass theory. Its reportable achievement is the mass-shape relation.

4.6 Interpretation

The formalism recovers a relation-level charged-lepton mass-shape pattern. The muon and tau MeV numbers appear only after using the known electron mass as a declared one-anchor ruler. The extraction is the ratio pattern, not a zero-anchor derivation of all three pole masses.

5 Result II: fixed-scope Standard-Model charge-lattice extraction

5.1 Target

The charge target is

$$Q_{SM}^{lat} = (\Lambda_{chg}, q, W_{chg}, T_{chg}, \sim_{\text{norm,sign}}).$$

It is a charge-lattice / hypercharge-ratio target, not a coupling-magnitude target. It is fixed-scope: the declared one-generation Standard-Model charge scope supplies the skeleton, charge witnesses, charged Yukawa operator neutrality, anomaly/tensor compatibility, conjugation, charge composition, closed redescription transport, and normalization/sign quotient.

The fixed scope does not supply the target hypercharge ratios as data. It supplies the admissibility problem: role skeleton, witness conditions, Yukawa neutrality where structurally licensed, anomaly/tensor compatibility, conjugation, charge composition, closed redescription transport, and normalization/sign quotienting. The extracted object is the unique normalized ratio class that survives those constraints inside the declared scope.

In compressed form, the charged-sector neutrality constraints are

$$Y_Q + Y_H - Y_u = 0, \quad Y_Q - Y_H - Y_d = 0, \quad Y_L - Y_H - Y_e = 0,$$

together with the anomaly-ledger constraints and admissible normalization/sign quotient. The displayed tuple is the normalized survivor of this constraint system, not an empirical charge table inserted as data.

The novelty claimed here is not the invention of new hypercharge algebra. It is the reportability result that, after the standalone structural envelope is fixed, the charge-lattice ratio class is the unique quotient-level survivor under the declared admissibility constraints.

5.2 Extracted quotient class

The extracted standing object is the hypercharge-ratio / lattice class

$$[Y_Q : Y_u : Y_d : Y_L : Y_e : Y_H]_{\text{norm,sign}} = [1 : 4 : -2 : -3 : -6 : 3]_{\text{norm,sign}}.$$

This quotient/lattice class is the extracted object. The coordinate tuple is its normalized presentation, not a primitive numerical constant list.

5.3 Normalized hypercharge presentation

Throughout the CS6A result, the convention is

$$Q = T^3 + Y.$$

This differs by a factor of two from the common convention $Q = T^3 + Y/2$. Under the primitive-unit convention $Y_Q = 1/6$, the normalized presentation of the quotient/lattice class is

$$Y_Q = \frac{1}{6}, \quad Y_u = \frac{2}{3}, \quad Y_d = -\frac{1}{3}, \quad Y_L = -\frac{1}{2}, \quad Y_e = -1, \quad Y_H = \frac{1}{2}.$$

5.4 Electric-charge presentation

Using $Q = T^3 + Y$, the electron-normalized presentation of the electric-charge lattice is

$$Q(u) = \frac{2}{3}, \quad Q(d) = -\frac{1}{3}, \quad Q(\nu) = 0, \quad Q(e) = -1.$$

The displayed fractions are lattice positions after normalization, not derivations of the electromagnetic coupling magnitude.

5.5 Status

The final CS7 status is

`FixedScopeChargeLatticeSingletonExtracted.`

5.6 Boundary

The charge-lattice result does not derive the electric coupling e , the fine-structure constant α , gauge couplings g_1, g_2, g_3 , the gauge group from no premises, the full Standard Model parameter table, or unrestricted BSM exclusion.

5.7 Interpretation

The formalism extracts the normalized Standard Model charge-lattice pattern: up quark $2/3$, down quark $-1/3$, electron -1 , and neutrino 0 , inside the declared fixed charge scope. It does not derive the strength of electromagnetism.

6 Structural support layer

The extraction results rely on a support layer that must remain distinct from numerical extraction. Each item below is certified structurally or as a ledger condition. None of them is, by itself, a numerical-value derivation.

6.1 Charge as standing-preserving invariant

Charge is not primitive in the ARC. It is a standing-preserving invariant required for coherent interaction accounting. This result supports the status

`ChargeInvariantCertified.`

6.2 Representation and charge-witness rigidity

Charge distinction is witness distinction. In compact form,

$$q \sim_{Ch} q' \iff \forall \alpha, W(q; \alpha) \sim W(q'; \alpha).$$

Raw labels, preferred normalizations, and unwitnessed distinctions collapse. The certified status is

`ChargeWitnessLedgerCertified.`

6.3 Abelian charge normalization

Abelian coordinates, charge ratios, normalization quotienting, lattice readiness, and $U(1)$ -like notation are licensed only as witness-preserving coordinate structures. Normalization is skin unless it changes the witness ledger. The status is

AbelianChargeNormalizationLicensed.

6.4 Anomaly as standing obstruction

A nonzero gauge anomaly obstruction is same-domain standing failure. This role-typing justifies anomaly neutrality as an admissibility condition rather than as an empirical add-on. The status is

AnomalyObstructionCertified.

6.5 Anomaly-ledger compression

Raw anomaly equations are not anomaly standing. They survive only as coordinates on admitted obstruction-neutrality ledgers over already-standing product, representation, and charge data. The status is

AnomalyLedgerCompressionCertified.

6.6 Anomaly–Yukawa equivalence

Under minimal chiral load and a single Higgs doublet, complete Yukawa realizability and anomaly freedom are equivalent tensor requirements. This certifies Yukawa operator existence as a cross-skin transport constraint, not optional mass-sector decoration:

anomaly freedom $\iff$ complete Yukawa realizability.

The status is

AnomalyYukawaEquivalenceCertified.

6.7 Higgs-like bridge carrier

The Higgs-like bridge survives only as canonical bridge carrier

$$B_D = (W_D^{br} / \sim_D^{br}, \text{Sig}^{br}, \beta_D^{red}, \rho_{br}).$$

The status is

BridgeCarrierRigidityCertified.

It does not derive the Higgs vev, Higgs mass, scalar self-coupling, weak scale, or weak mixing angle.

6.8 Yukawa topology readiness

Yukawa-like quantities survive only as coupling-witness structures over the canonical bridge carrier. Raw entries, texture choices, fitted values, mass matrices, and RG trajectories do not license coupling standing. The status is

YukawaTopologyReadinessCertified.

No numerical Yukawa eigenvalues are derived.

7 Obstruction result: ZPE quotient-descent obstruction

7.1 Target

The target is the restricted homogeneous metric-proportional vacuum sector V_0 . The object at issue is the absolute homogeneous matter-sector zero-point offset, not the full renormalized stress tensor in arbitrary curved spacetime.

7.2 Formal result

The CS6B report is

$$\rho_R \notin O_{\text{pred}}(V_0),$$

where $O_{\text{pred}}(V_0)$ is the class of quotient-invariant standalone predictive content in the restricted homogeneous sector.

7.3 Meaning

The absolute homogeneous zero-point offset fails quotient descent under arbitrary finite homogeneous shifts in the restricted sector. It is therefore not standalone predictive content of that sector.

7.4 Surviving physics

The obstruction does not erase vacuum physics. The ARC preserves:

- Casimir forces as relational response quantities;
- phase-transition energy differences as relational/dynamical comparisons;
- Λ_{eff} as a boundary-matched effective parameter;
- the naturalness and radiative-stability problem as unresolved.

7.5 Status

The final status is

ZPE quotient obstruction.

7.6 Boundary

CS6B does not derive the observed cosmological constant, explain dark energy, solve naturalness, deny Casimir forces, deny phase-transition vacuum differences, or replace QFT. It removes one invalid prediction-forming map; it does not solve the cosmological-constant problem.

7.7 Interpretation

The ARC does not deny vacuum effects. It states that the naive zero-point sum is not a valid standalone prediction of the cosmological constant. The cosmological-constant problem remains, but the familiar cutoff-sum comparison is not a reportable prediction map.

8 Structural reconstruction result: neutrino oscillation

8.1 Boundary kernel

Within the declared finite, coherent, noncommuting PMNS source–propagation–detection scope, the same-scope reconstruction is the boundary kernel

$$K_{\beta\alpha}(t) = \langle \nu_\beta | U(t) | \nu_\alpha \rangle.$$

The status is

`OscillationReconstructionCertified.`

8.2 Invariant witnesses

The oscillation-regime invariant witness classes are

$$\Delta m_{ij}^2$$

and

$$U_{\alpha i} U_{\beta i}^* U_{\alpha j}^* U_{\beta j}.$$

Their status is

`NeutrinoInvariantWitnessCertified.`

The values of these witnesses are not derived by CS6C.

8.3 Interior flavor totalization obstruction

Weak flavor is boundary-defined:

$$T_\alpha : X_\nu \rightharpoonup F.$$

It cannot be same-scope totalized over the propagation interior when the propagation operator does not preserve the flavor projectors. The status is

`InteriorFlavorTotalizationObstructed.`

8.4 Boundary

CS6C does not derive PMNS matrix values, mass-splitting numerical values, CP phase, absolute neutrino mass scale, sterile-sector exclusion, or any modification of oscillation phenomenology. It also does not deny flavor at weak-interaction production and detection. It denies only globally transported interior weak-flavor identity under noncommuting propagation.

8.5 Interpretation

Flavor is real at production and detection. It is not carried as a fixed internal label during propagation when the propagation dynamics do not preserve the flavor projectors. The correct object is the boundary-to-boundary oscillation kernel.

9 Deferred numerical frontier

The deferred ledger is an achievement of reportability, not a weakness. It states exactly where the present corpus stops.

9.1 Coupling constants

The following remain deferred:

$$e, \quad \alpha, \quad g_1, g_2, g_3.$$

The charge lattice derives charge ratios and lattice positions, not coupling magnitudes.

9.2 Higgs and electroweak scale

The following remain deferred:

$$v, \quad m_H, \quad \lambda_H, \quad m_W, \quad m_Z, \quad \sin^2 \theta_W.$$

Bridge carrier standing does not derive scale values.

9.3 Yukawa eigenvalues

Numerical Yukawa eigenvalues remain deferred. Yukawa topology readiness is not numerical Yukawa extraction.

9.4 Fermion masses

The charged-lepton relation-level shape is extracted. Exact charged-lepton pole masses, zero-anchor charged-lepton scale, quark masses, and neutrino absolute masses remain deferred.

9.5 Mixing matrices

CKM and PMNS numerical coordinates remain deferred. Invariant classes may be certified; numerical values are not extracted.

9.6 Cosmological values

The following are not derived:

$$\Lambda_{\text{eff}}, \quad \rho_\Lambda, \quad H_0.$$

ZPE obstruction is not cosmological-value extraction.

10 Master reportability ledger

Table 2 gives the compact final ledger. The accompanying registry materials include the corresponding machine-readable CSV ledger.

Table 2: Master reportability ledger.

ID	Target	Final status	Reportable form and blocked promotions
CL-1	Charged-lepton mass-shape	Extracted	Relation-level ratio [1 : 206.7703159727 : 3477.4728371047]. Blocks exact pole masses and zero-anchor scale.
CL-2	One-anchor charged-lepton display	Presentation only	MeV display using empirical m_e . Blocks zero-anchor derivation.

CL-3	Exact charged-lepton pole masses	Deferred	No reportable value; protected trace and pole-boundary certificates remain absent.
SMQ-1	Fixed-scope Standard-Model charge lattice	Extracted	Quotient/lattice class $[1 : 4 : -2 : -3 : -6 : 3]_{\text{norm,sign}}$. Blocks promotion to e, α, g_i and full parameter-table closure.
SMQ-2	Normalized hypercharge table	Presentation only	Normalized display of quotient class; not raw numerical constants.
SMQ-3	e, α, g_i	Deferred	No coupling-magnitude value is derived from the charge lattice.
HBR-1	Higgs-like bridge carrier	Certified structural	Canonical bridge carrier B_D . Blocks Higgs vev, Higgs mass, and weak-scale claims.
HBR-2	Higgs / electroweak scale	Deferred	No reportable value; bridge carrier does not determine scale values.
YUK-1	Yukawa topology	Certified structural	Coupling-witness topology. Blocks numerical Yukawa eigenvalue claims.
YUK-2	Yukawa eigenvalues	Deferred	No reportable value; topology is not eigenvalue extraction.
ZPE-1	Absolute homogeneous ZPE offset	Certified obstruction	Nonpredictivity / quotient obstruction. Blocks cosmological-constant solution claims.
ZPE-2	Cosmological constant value	Deferred	Boundary-matched parameter; not derived by ZPE obstruction.
OSC-1	Boundary kernel $K_{\beta\alpha}$	Certified structural	PMNS boundary-kernel reconstruction. Blocks PMNS numerical value claims.
OSC-2	PMNS invariant witnesses	Certified structural	Δm_{ij}^2 and rephasing-invariant products as witness classes; numerical values deferred.
OSC-3	Transported weak flavor	Certified obstruction	Interior flavor totalization blocked. Boundary flavor is preserved.
GN-9	Full cross-sector aggregate	Blocked	Master route undefined. Blocks universal derivation from pooled nodes.

11 Forbidden-promotion theorem summary

11.1 Theorem 1: Status non-amplification

No target may be reported above its certified status for its declared scope. Compatibility with stronger-looking structures does not discharge an unresolved blocker.

11.2 Theorem 2: Presentation / extraction separation

Normalized displays, one-anchor displays, and formula displays are not extracted constants unless the quotient-level target itself has extraction status.

11.3 Theorem 3: Structural / numerical separation

Certified structural objects do not imply numerical values. A bridge carrier is not a Higgs vev. Yukawa topology is not a Yukawa eigenvalue table. PMNS invariant witness certification is not PMNS numerical extraction.

11.4 Theorem 4: Obstruction / explanation separation

Obstruction certificates block routes. They do not automatically explain target values. ZPE quotient obstruction does not solve dark energy, and interior flavor totalization obstruction does not derive PMNS values.

11.5 Theorem 5: Master-network nondefinition

The full cross-sector aggregation is not a singleton route. Pooled attractive nodes do not become a universal derivation in the absence of a single target ledger and complete blocker discharge.

11.6 Theorem 6: CS synthesis theorem

The Cross-Sector ARC establishes exactly the ledger categories reported in this synthesis and no stronger unregistered claims.

12 Why the result is evidence for AASC

12.1 The evidential point

A constraint formalism gains evidential force when the same admissibility discipline isolates more than one independent quotient target without target leakage. The CS ARC supplies two distinct extracted structures:

charged-lepton mass-shape

and

Standard-Model charge-lattice.

These are not the same sector. Isolating both under one reportability discipline is materially stronger than fitting one isolated pattern, because the same constraint discipline survives two different quotient targets while refusing stronger unlicensed promotions.

12.2 Cross-sector extractive strength

The two positive results are distinct in kind:

charged-lepton mass ratios $\neq$ Standard-Model charge ratios.

The former concerns a mass-shape relation. The latter concerns charge-lattice / hypercharge-ratio structure. AASC also identifies which neighboring claims do not yet have standing. This combination of positive extraction and controlled refusal is the strongest current evidence supplied by the CS ARC.

12.3 Prospective stronger test

A new unknown numerical value later confirmed would provide stronger evidence. A realistic forward route is a Neutrino Numerical-Readiness ARC targeting discrete or interval-level outputs such as mass ordering, CP-phase class, θ_{23} octant, or absolute-mass bounds. The present synthesis motivates that next program without claiming it has already been achieved.

12.4 Final evidential claim

The CS ARC does not prove the entire AASC formalism correct. It supplies cross-sector evidence that the formalism isolates nontrivial quotient-level structure: one charged-lepton mass-shape pattern and one Standard-Model charge-lattice pattern, while blocking several tempting but unlicensed numerical claims.

13 Conclusion

13.1 What was achieved

The Cross-Sector ARC established two extracted quotient-level targets, several certified structural and invariant targets, two dedicated obstruction branches, full-network nondefinition as a blocked master-route result, and a deferred numerical frontier.

The two extracted targets are:

1. charged-lepton relation-level mass-shape;
2. fixed-scope Standard-Model charge-lattice / hypercharge-ratio class.

The certified structural and invariant targets include the charge witness ledger, Abelian normalization, anomaly-standing obstruction, anomaly-ledger compression, anomaly–Yukawa equivalence, bridge carrier rigidity, Yukawa topology readiness, neutrino boundary-kernel reconstruction, and PMNS invariant witness classes. The dedicated obstruction branches are ZPE quotient-descent obstruction and interior weak-flavor totalization obstruction.

13.2 What was not achieved

The ARC did not derive the full Standard Model parameter table. It did not derive coupling constants, Higgs/electroweak scale, numerical Yukawa eigenvalues, CKM/PMNS values, exact charged-lepton pole masses, zero-anchor charged-lepton mass scale, cosmological constant value, or BSM impossibility.

13.3 Final posture

The Cross-Sector ARC is therefore best read as a reportability-controlled proof of cross-sector quotient extraction and obstruction discipline, not as an unrestricted numerical completion of the Standard Model.

A Final registry schema

The accompanying registry materials include a CSV ledger using the following schema:

<code>object_id</code>	<code>target</code>	<code>source</code>	<code>scope</code>	<code>value_class</code>
<code>primary_status</code>	<code>reportable_form</code>	<code>blocked_promotions</code>	<code>blockers</code>	<code>next_certificate</code>

This schema enforces the difference between extracted values, normalized presentations, structural objects, obstruction certificates, deferred targets, and blocked routes.

B Paper-to-result map

Table 3: Paper-to-result map.

Paper	Function	Final use
CS1	Candidate admission and claim-standing discipline	Registry discipline and anti-calibration firewall
CS2	Same-target spectral/residue/mixing audit	Charged-lepton relation target fixation
CS3	Charge/anomaly/residual typing	Charge/anomaly structural support and normalization control
CS4	Hierarchy/bridge/scale audit	No absolute-scale promotion
CS5	Global compatibility and no-amplification	Relation-only CS6 admission; full-network nondefinition
CS5.5	Non-CL target search	Charge lattice selected; ZPE and neutrino branches retained
CS6A	Fixed-scope charge-lattice extraction	Fixed-scope charge-lattice singleton extracted
CS6B	ZPE quotient obstruction	ZPE quotient obstruction certified
CS6C	Neutrino boundary-kernel reconstruction	Oscillation reconstruction and neutrino invariant witnesses certified
CS7	Final registry	Status ledger and forbidden-promotion control

C Forbidden promotion table

Table 4: Expanded forbidden-promotion table.

Forbidden promotion	Correct status
Charged-lepton relation ratio to exact pole masses	Exact pole closure deferred.
One-anchor charged-lepton display to zero-anchor derivation	Presentation only / OneAnchorMetricDisplay.
Charge lattice to e, α, g_i	Coupling magnitude deferred.
Normalized hypercharge table to raw numerical constants	NormalizedPresentation only.
Bridge carrier to Higgs vev	Higgs/electroweak scale deferred.
Yukawa topology to Yukawa eigenvalues	Numerical Yukawa values deferred.
Anomaly obstruction to mass prediction	Forbidden; anomaly standing is structural/obstruction support.
ZPE obstruction to cosmological-constant solution	Obstruction only; naturalness unresolved.
PMNS invariant witness to numerical PMNS values	Numerical oscillation parameters deferred.

Table 4: Expanded forbidden-promotion table.

Forbidden promotion	Correct status
Boundary flavor trace to transported interior flavor	Interior totalization obstructed.
Global compatibility to singleton extraction	Singleton requires explicit extraction certificate.
Full network aggregation to master derivation	Blocked by GlobalNetworkNotDefined.

D Clean final claims list

The final reportable claims are:

1. The charged-lepton relation-level mass-shape ratio is algebraically extracted as $[1 : 206.7703159727 : 3477.4728371047]$.
2. The muon and tau MeV readouts are one-anchor displays using the empirical electron mass.
3. The fixed-scope one-generation Standard-Model charge-lattice / hypercharge-ratio class is extracted as $[1 : 4 : -2 : -3 : -6 : 3]_{\text{norm,sign}}$.
4. The normalized hypercharge and electric-charge tables are presentations of that quotient/lattice class, not coupling-value derivations.
5. The absolute homogeneous zero-point vacuum-energy offset is not standalone predictive content in the restricted homogeneous sector.
6. Neutrino oscillation in the declared finite PMNS scope is certified as boundary-kernel reconstruction, with invariant witnesses Δm_{ij}^2 and rephasing-invariant PMNS products.
7. Globally transported interior weak-flavor identity is obstructed in the noncommuting PMNS scope.
8. The Higgs-like bridge carrier and Yukawa topology readiness are certified structurally, not numerically.
9. The full cross-sector aggregation remains blocked as a master singleton route.

E Deferred target ledger

The deferred target ledger contains:

- e, α, g_1, g_2, g_3 : coupling exactness certificate needed;
- Higgs/electroweak values $v, m_H, \lambda_H, m_W, m_Z, \sin^2 \theta_W$: scale certificate needed;
- numerical Yukawa eigenvalues: coupling-to-mass extraction certificate needed;
- exact charged-lepton pole masses: protected trace-remainder certificate needed;
- zero-anchor charged-lepton scale: scale-carrier certificate needed;
- CKM/PMNS numerical values: mixing-value extraction certificate needed;
- cosmological values $\Lambda_{\text{eff}}, \rho_\Lambda, H_0$: cosmological matching/transport certificate needed;
- full Standard Model parameter table: multiple downstream certificates needed.

F Transition to the Neutrino Numerical-Readiness ARC

The next forward-prediction attempt is naturally located in the neutrino sector because CS6C already isolates the admissible invariant witness classes without deriving numerical values. Candidate targets for a prediction-lock program include mass ordering, CP-phase class, θ_{23} octant, or absolute-mass bounds. Any such ARC would require pre-registration of the target, exclusion of empirical target leakage, a frozen derivation path, and a specified experimental comparison channel before confirmation.

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